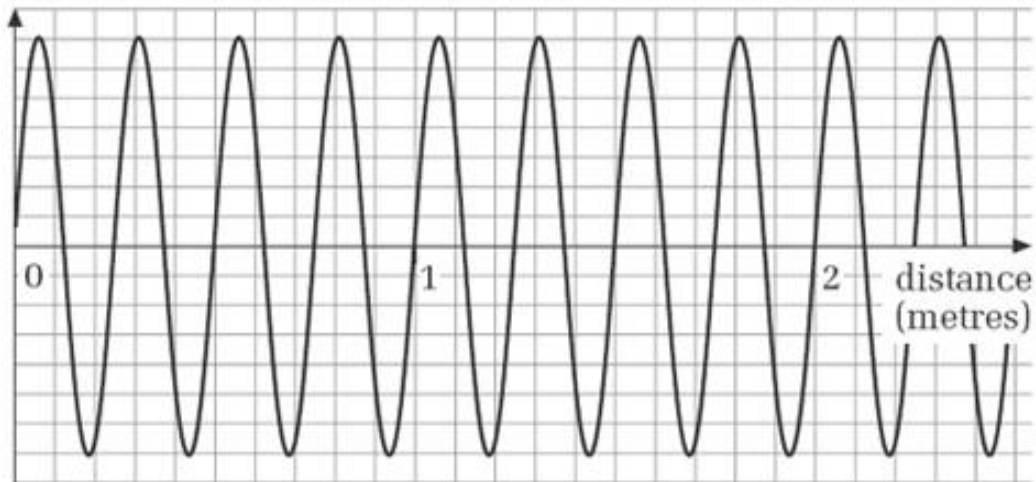
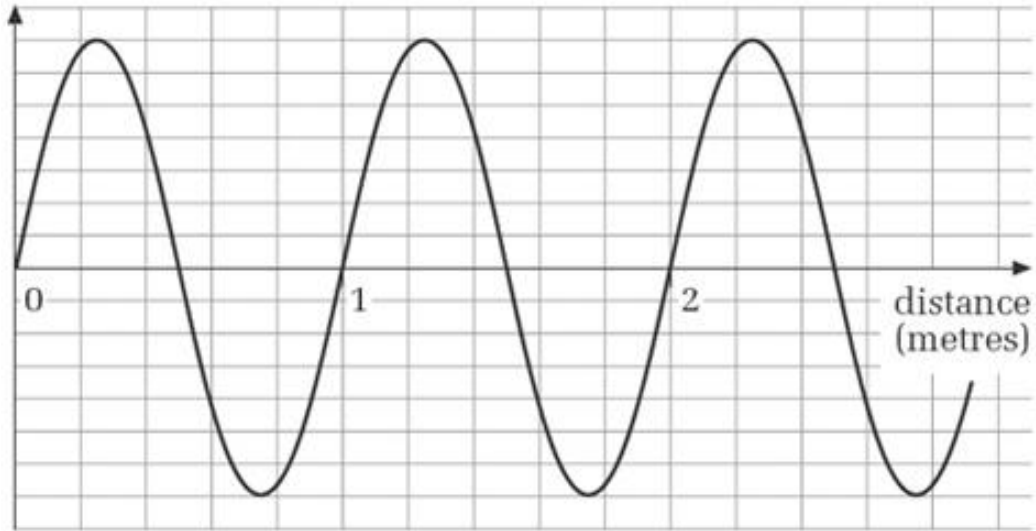
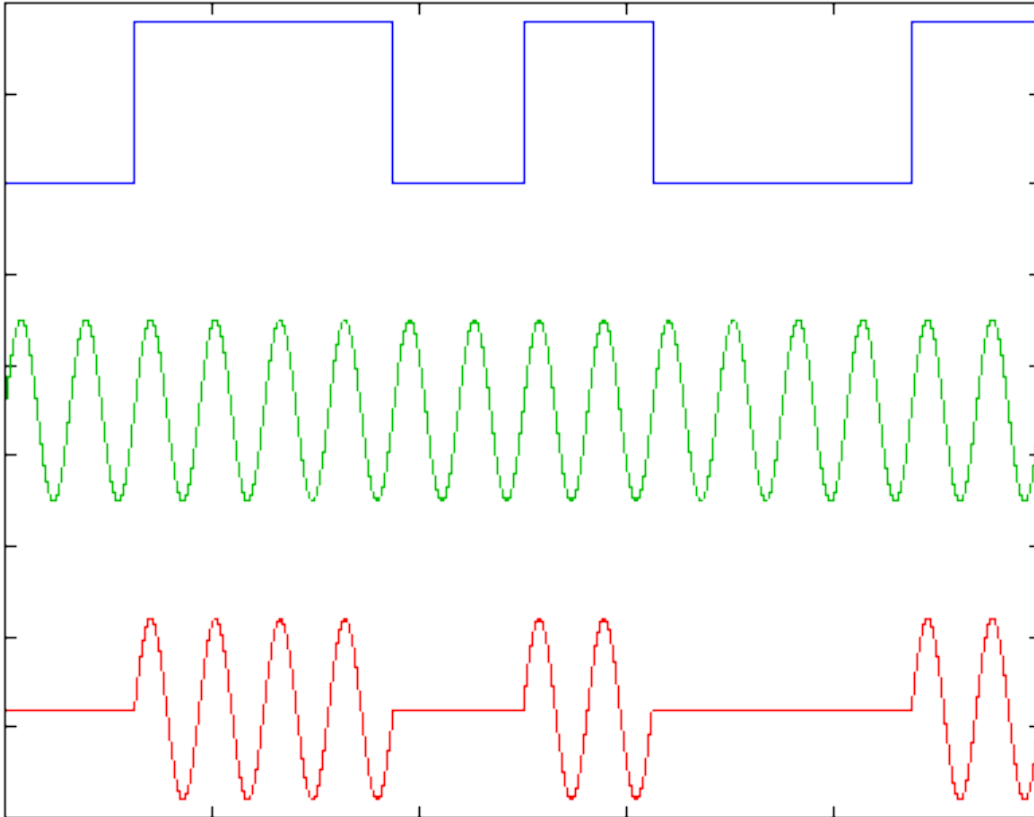


Carrier Signal (Sinusoidal)



Amplitude Modulation



Modulation is the process of merging two signals into a third signal with desirable properties of both.

Amplitude modulation involves multiplication of the amplitudes of the two signals.

A concrete example of AM

Modulation results in radio waves that can propagate over long distances and can carry voice or other information.

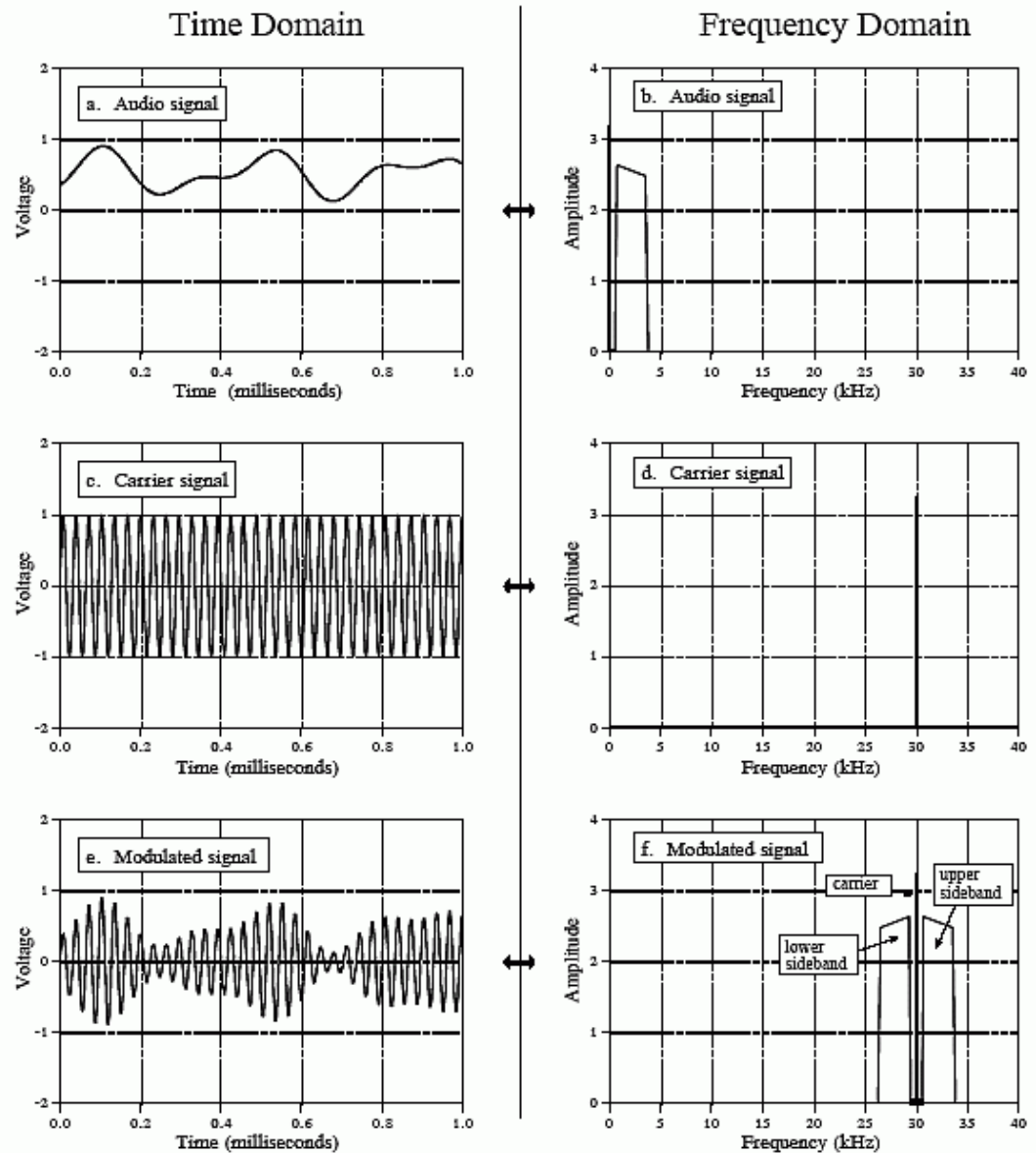
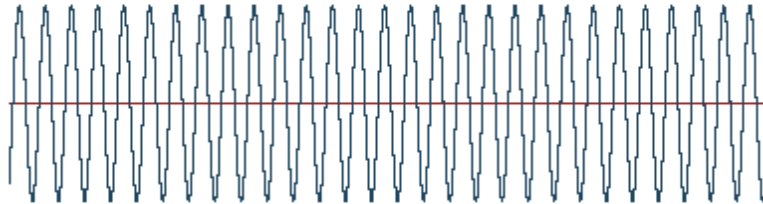


FIGURE 10-14 Amplitude modulation. In the time domain, amplitude modulation is achieved by multiplying the audio signal, (a), by the carrier signal, (c), to produce the modulated signal, (e). Since multiplication in the time domain corresponds to convolution in the frequency domain, the spectrum of the modulated signal is the spectrum of the audio signal shifted to the frequency of the carrier.

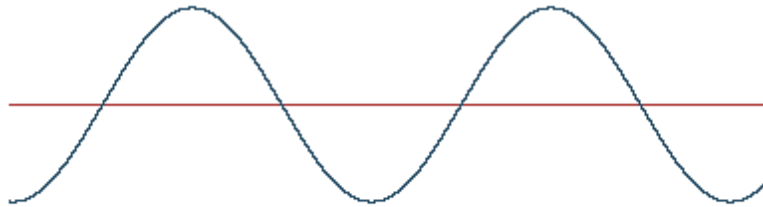
Amplitude Modulation

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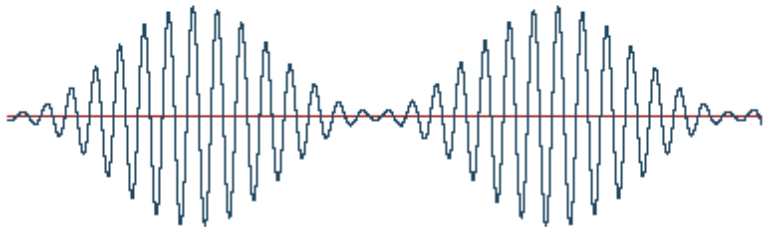
Carrier



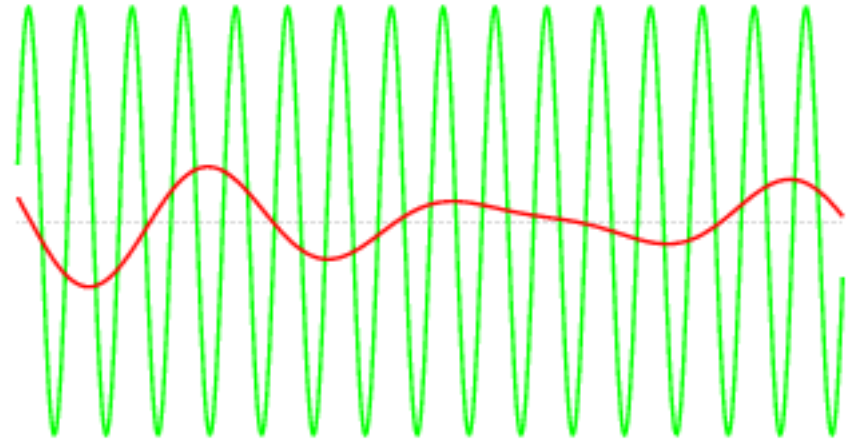
Modulating Wave



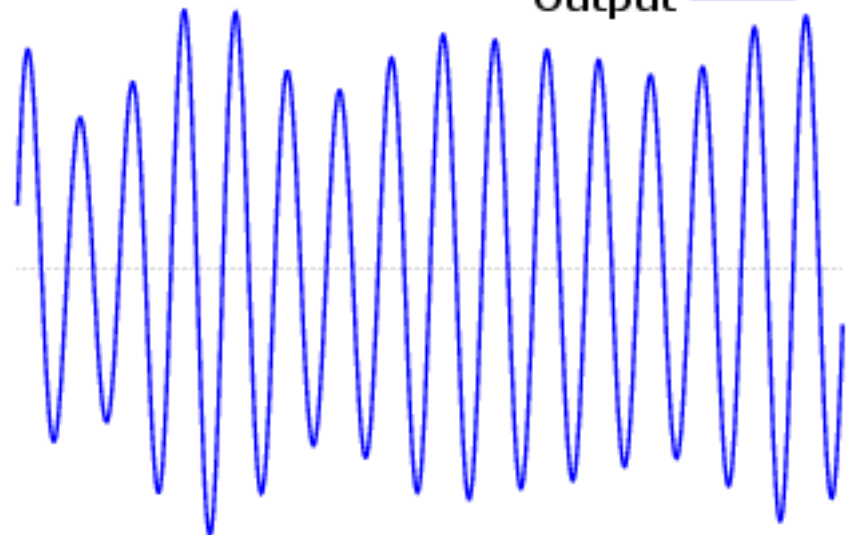
Modulated Result



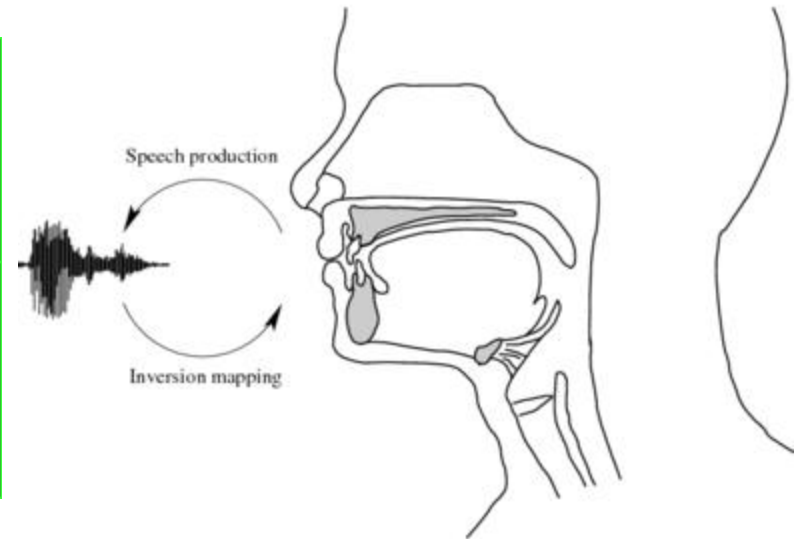
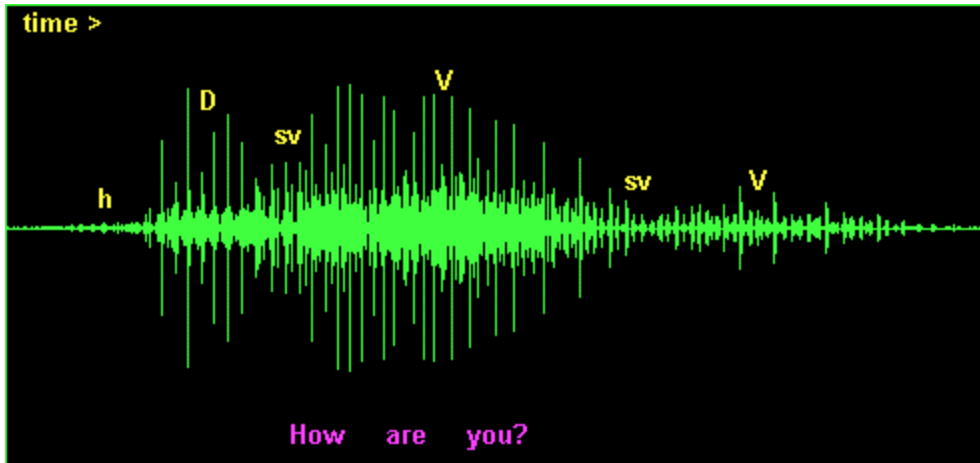
Carrier —
Signal —



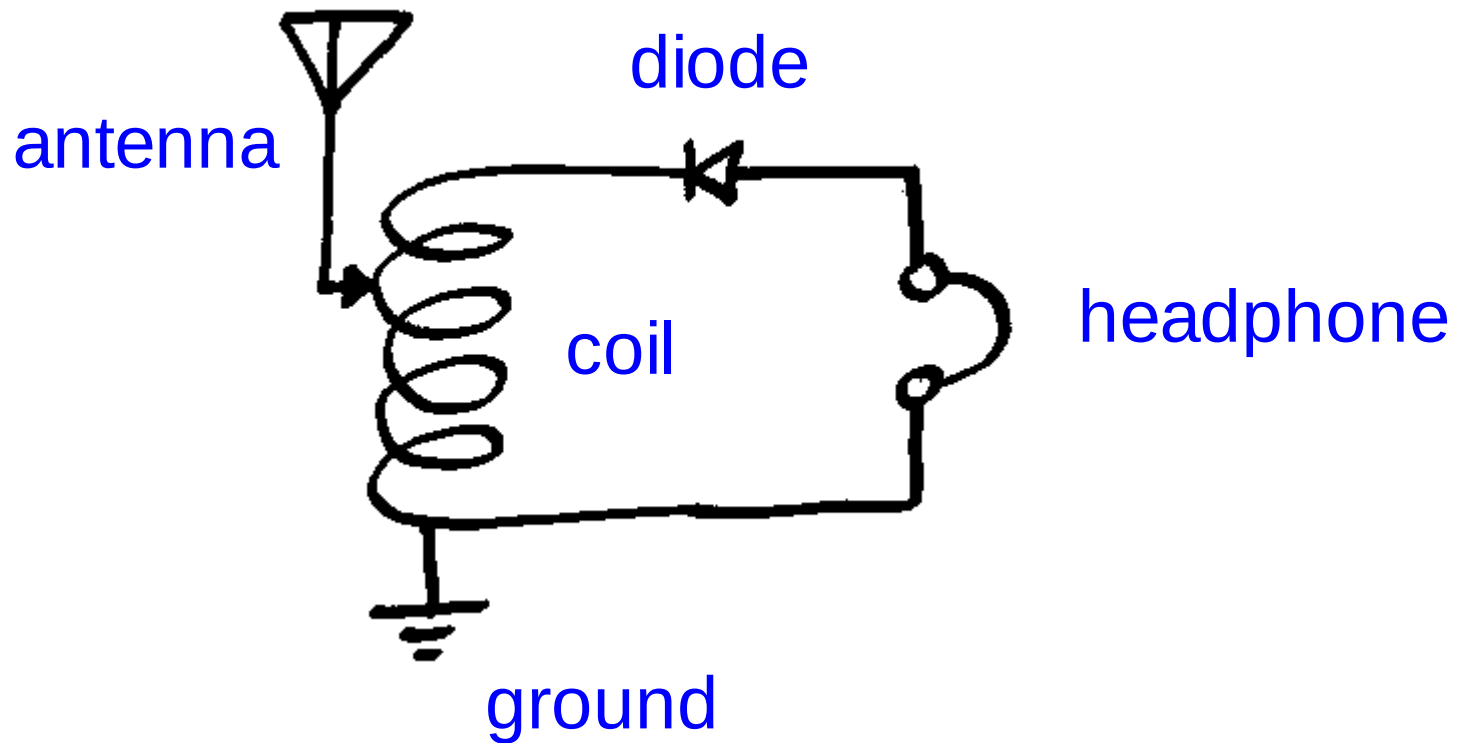
Output —



Speech Signal



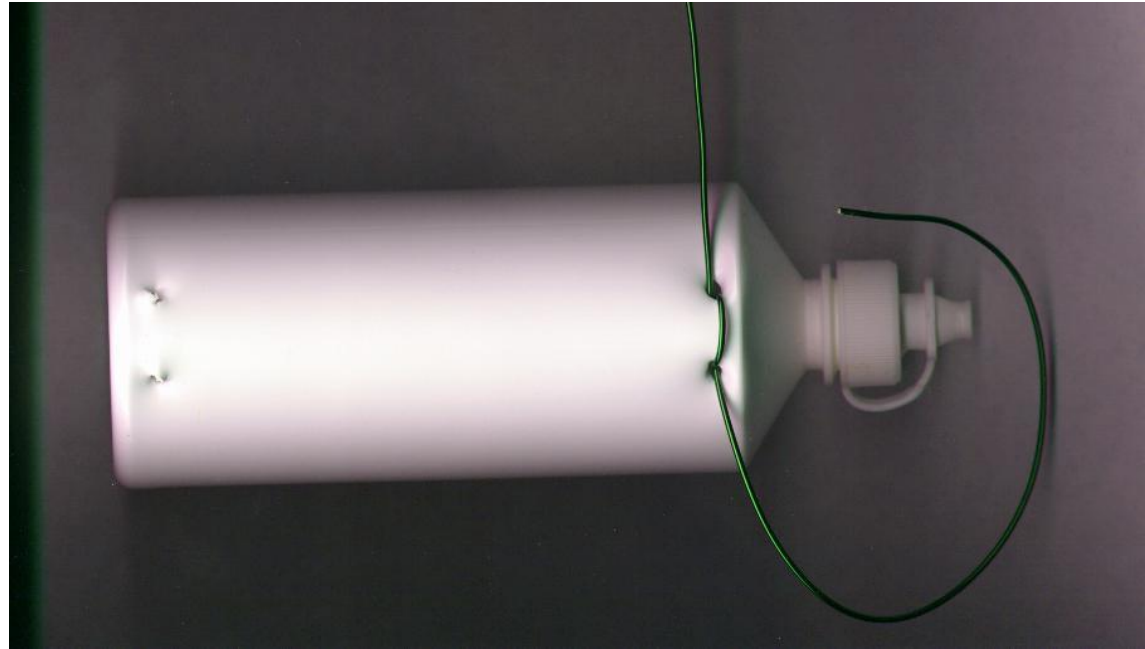
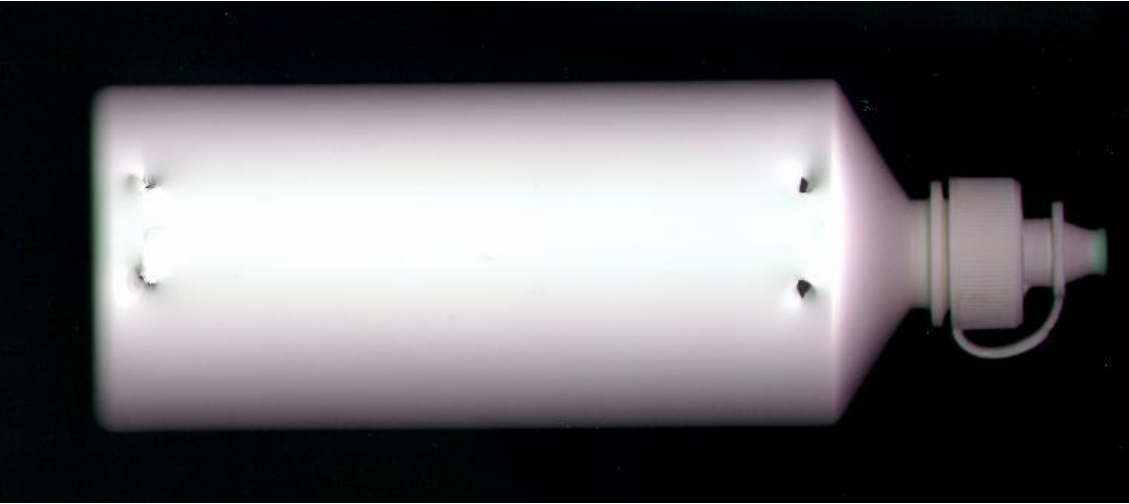
Crystal Radio (A Schematic)



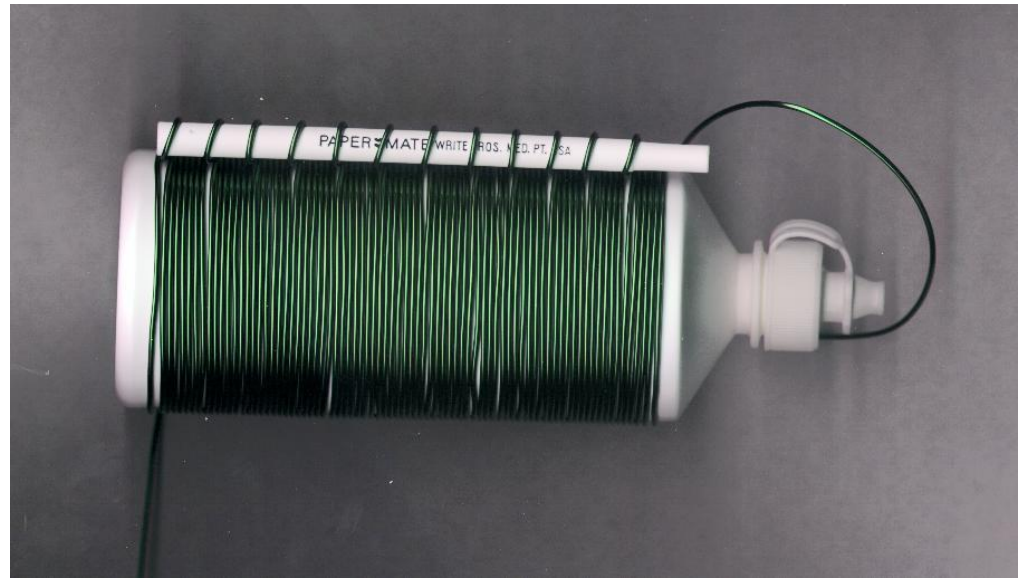
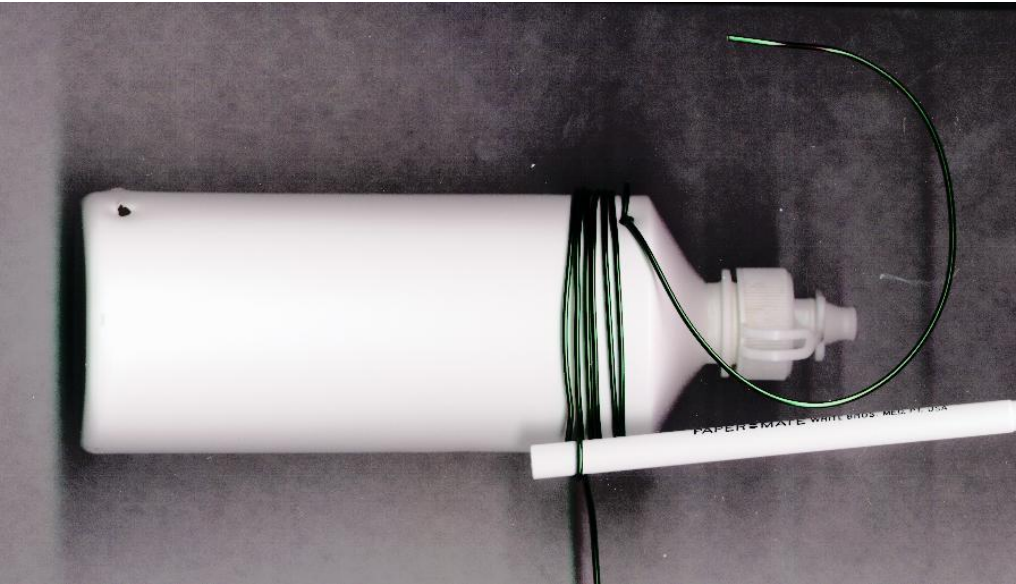
A Foxhole Radio



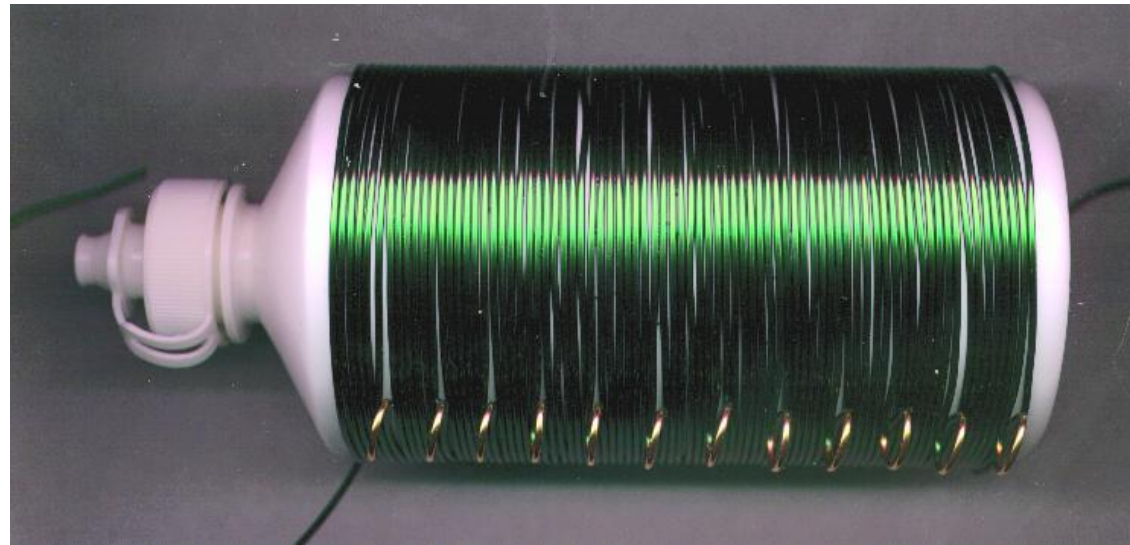
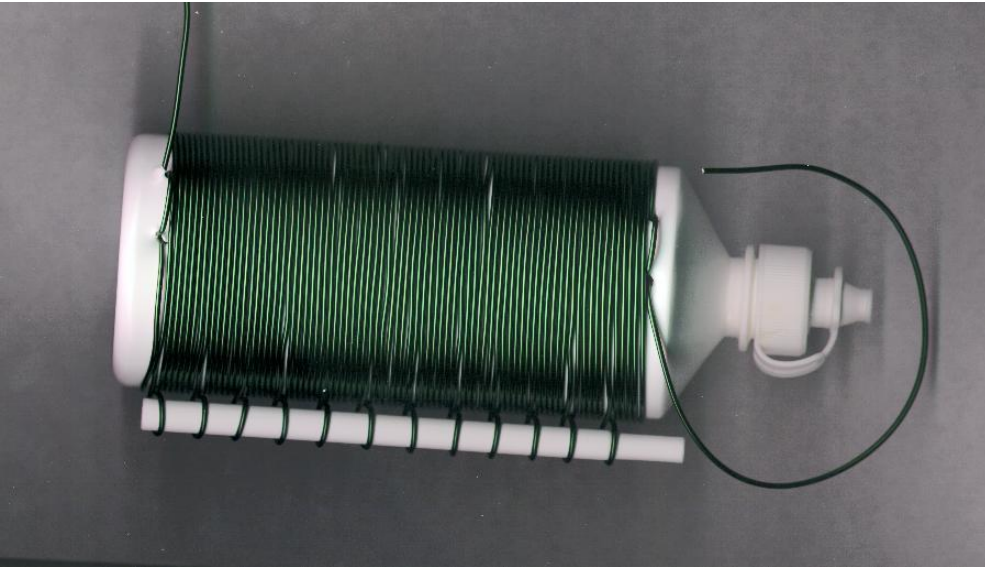
Making a Foxhole Radio



Making a Foxhole Radio



Making a Foxhole Radio



Making a Foxhole Radio

